and enabling immediate decision-making. Real-time analytics and AI-based data fusion empower drones to identify targets, navigate complex terrains, and respond to threats instantly.

- *Advanced communication and networking systems*
  Incorporating 5G, quantum-encrypted links, and satellite-based communication, fourth-generation drones ensure secure, high-speed data transfer between multiple units and command centers. These technologies facilitate beyond-line-of-sight (BLOS) operations, even in contested or GPS-denied environments.
- *Enhanced stealth and survivability*
  Designed using low-observable (LO) materials, adaptive camouflage, and reduced radar and infrared signatures, fourth-generation drones are optimized for survivability in high-threat zones. Their stealth characteristics allow deep-penetration missions and long-endurance reconnaissance in hostile territories.
- *Modular and scalable design*
  These drones employ a modular architecture, allowing easy customization for various roles – such as ISR (intelligence, surveillance, reconnaissance), EW (electronic warfare), cargo delivery, or combat missions. Scalability also enables integration with emerging technologies such as directed-energy weapons or hypersonic propulsion.
- *Autonomy and human–machine teaming (HMT)*
  Fourth-generation UAVs operate autonomously yet remain connected through human–machine teaming interfaces, allowing commanders to supervise missions without direct piloting. This feature supports loyal wingman operations, where drones fly alongside manned aircraft to extend situational awareness and combat capability.